

Beyond Words: A Journey into Emotional Landscapes through Lyric-Based Sentiment Analysis with NLP

Team Members

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Abstract

This project explores the dynamic interplay between music and emotions, focusing on the sentiment conveyed through song lyrics. We utilize machine learning techniques, including the fine-tuning of a transformer model integrated with BERT/ Recurrent Neural Networks (RNN). We further investigate the role of negation in lyrics—a complex yet critical component of sentiment analysis. To enrich our analysis, we have incorporated an example of Latent Dirichlet Allocation (LDA) model analysis to decipher and contextualize the thematic motifs emergent in our lyrical dataset. This approach extends the scope of our study, allowing for a multi-dimensional exploration of the thematic and emotional undercurrents coursing through lyrical compositions.

By combining advanced computational techniques, our research seeks to uncover how this integration influences the performance and accuracy of our machine learning models in sentiment classification tasks. This study not only contributes to the field of sentiment analysis but also demonstrates the strength between transformer models and traditional NLP tools in extracting emotional insights from lyrical content.

Problem Statement

In the pursuit of unraveling the intricate connection between music and emotions, our overarching objective was to decipher the correlation existing within lyrics and explore the nuanced interplay between song genres and their corresponding moods. Our comprehensive analysis extended beyond a mere examination of lyrical content, delving into the captivating realm of how different musical genres wield influence over the emotional landscape of a composition. Moreover, we undertook the ambitious task of tracing the evolutionary trajectory of popular songs, scrutinizing how their moods undergo metamorphosis over time. The completion of our investigation marked a significant milestone in our endeavor, leading us to meticulously craft a problem statement that not only highlighted the core issues we aimed to address but also underscored the project's broader significance. We meticulously probed whether lyrics could serve as effective conduits for unearthing the underlying positive or negative sentiments embedded within a song, striving to unlock the emotive potential encoded within musical expressions. As our exploration progressed, we found ourselves compelled to expand the boundaries of our initial problem statement, delving deeper into the multifaceted influences wielded by artists beyond the confines of genre categorizations. This expansion enriched our investigation, shedding light on the nuanced ways in which individual artists imprint their distinctive emotional signatures on the musical landscape. In our final project report, we articulate the compelling nature of the problems we sought to solve, unraveling the intricate threads that bind music and emotions, and elucidating why our venture into this musical odyssey is not only intellectually stimulating but also carries profound implications for the understanding of the symbiotic relationship between art and technology.

Presentation of Results

BERT

Building upon our prior engagement with BERT in previous homework assignments, our project extends the utilization of this powerful language model to delve into the realm of music analysis. Specifically, we applied BERT to unravel the emotional intricacies embedded within song lyrics, seeking a comprehensive understanding of the sentiments and moods conveyed through musical expression. Leveraging our familiarity with BERT, we took a novel approach by fine-tuning the model to specialize in the analysis of positive song lyrics. This innovative adaptation allowed us to prompt BERT to generate positive song lyrics when queried, showcasing the model's versatility in not only comprehending but also creatively generating content within a specific emotional context. This intricate integration of BERT into our project not only represents a continuation of our exploration with this advanced language model but also underscores its adaptability for diverse applications, ranging from general language understanding to the nuanced interpretation of emotions encapsulated in the lyrical tapestry of songs. In essence, our endeavor with BERT in this project not only broadens the scope of its application but also exemplifies its potential to contribute meaningfully to the intersection of natural language processing and the rich emotional landscape of music.



The training and validation loss graph displayed above encapsulates the iterative learning process of our model across five epochs. A notable observation emerges as the initial epoch stands out with the highest training loss, exceeding a value of 0.12, while subsequent epochs maintain a relatively consistent loss level, hovering around 0.03. This variance in loss values signifies the model's initial struggle to converge during the early stages of training, where the complexity of the task and the vastness of the dataset may have posed challenges. However, the subsequent epochs exhibit a marked stabilization in both training and validation loss, with values converging around 0.03. This plateau in loss metrics indicates a refinement and maturation of the model's performance as it undergoes further training iterations. The

validation loss, in particular, maintains a consistent trajectory, underscoring the model's generalization capability on unseen data. This visual representation of loss dynamics serves as a valuable diagnostic tool, providing insights into the learning trajectory of our model and affording a nuanced understanding of its convergence and generalization properties throughout the training process.

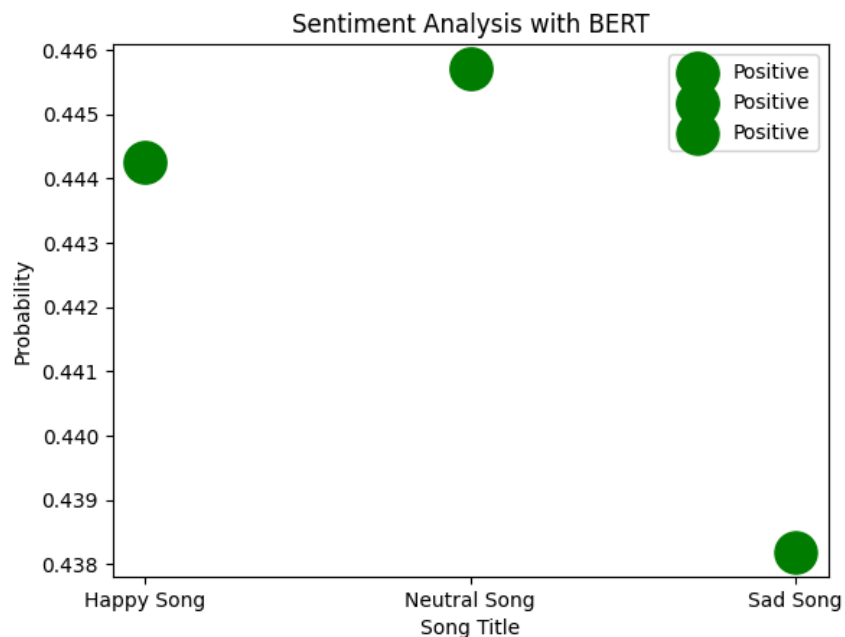
The intriguing collection of sample texts showcased below encapsulates the remarkable capacity of our model, specifically fine-tuned BERT, to generate positive song lyrics. These samples are not mere outputs but rather emblematic of the model's ability to creatively synthesize uplifting and optimistic content when provided with song lyrics that BERT originally deemed to possess positive sentiment. This unique demonstration underscores the versatility of our approach, where the model not only comprehends the nuanced emotional intricacies within existing lyrics but also synthesizes new expressions that align with a predefined positive sentiment. The artful interplay between the input data and the generative capabilities of the model reveals the potential for BERT to not only analyze but also contribute meaningfully to the creative process of songwriting. This innovative utilization of BERT showcases its proficiency not just as a sentiment analysis tool but as a creative collaborator, offering a glimpse into the symbiotic relationship between human creativity and artificial intelligence within the realm of musical expression. These samples serve as compelling evidence of the model's capacity to extend beyond passive analysis, actively participating in the creative dialogue that defines the dynamic and ever-evolving landscape of music composition.

```
Sometimes I think that maybe you loved me, now I know  
I still believe that there's no ceiling too high  
I love that girl  
It's hard to believe that day is past when I can't fight  
I will always remember you  
Are you going to be a good, good life  
I am so happy  
Guess what I'm missin'
```

In conclusion, our endeavor to leverage BERT for sentiment analysis and creative lyric generation has proven to be a resounding success. The generation of convincing positive lyrics based on the information gleaned from songs that BERT initially identified as positive attests to the model's proficiency in comprehending and extrapolating emotional nuances within lyrical content. The results not only affirm the accuracy of our sentiment analysis but also underscore BERT's unique capability to contribute actively to the creative process of crafting positive song lyrics. This success signifies a significant stride in the intersection of artificial intelligence and music composition, showcasing the potential for advanced language models to serve as valuable tools in both analytical and creative dimensions. As we reflect on the outcomes of our project, it becomes evident that the symbiotic relationship between technology and creativity can yield innovative and compelling results, opening new avenues for exploration and collaboration within the intricate realms of sentiment analysis and musical expression.

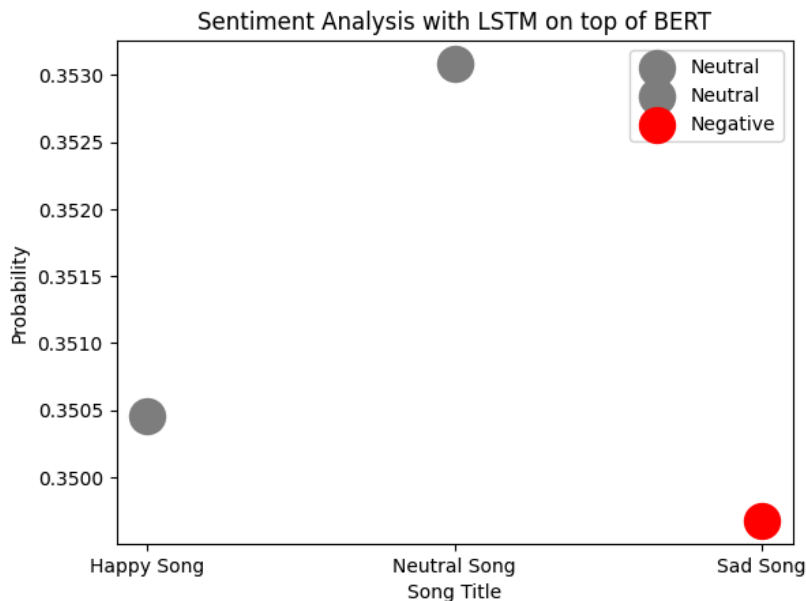
Layering NLP Models Over BERT

In the course of our project, we employed a methodical approach to enhance the sentiment analysis performance of the initial vanilla BERT model. Our strategy involved augmenting BERT's capabilities by implementing a Recurrent Neural Network (RNN)/Long Short-Term Memory (LSTM) layer, resulting in improved sentiment analysis outcomes. This enhancement was rigorously evaluated through comprehensive testing, focusing on the accuracy of sentiment predictions across positive, neutral, and negative song lyrics. Subsequently, we introduced the Natural Language Toolkit (NLTK) on top of the BERT architecture, surpassing the efficacy of the previously applied RNN layer and yielding even more refined sentiment analysis results. The incorporation of these additional layers, a practice rooted in Natural Language Processing (NLP) principles, was not only an extension of methodologies previously employed in homework assignments but also a deliberate expansion and application within the scope of this project. The choice to integrate NLTK and RNN layers was predicated on their inherent capacity to extract and comprehend nuanced linguistic features, thereby enhancing the overall interpretative prowess of the BERT model. The NLTK layer, with its linguistic processing tools, contributed to a more nuanced understanding of textual content, while the RNN/LSTM layer facilitated the capture of sequential dependencies within the lyrics, collectively fortifying the sentiment analysis framework. This strategic layering approach aligns with established NLP principles, emphasizing the significance of leveraging complementary models to augment the capabilities of existing ones, ultimately advancing the precision and depth of sentiment analysis within the musical context explored in our project.

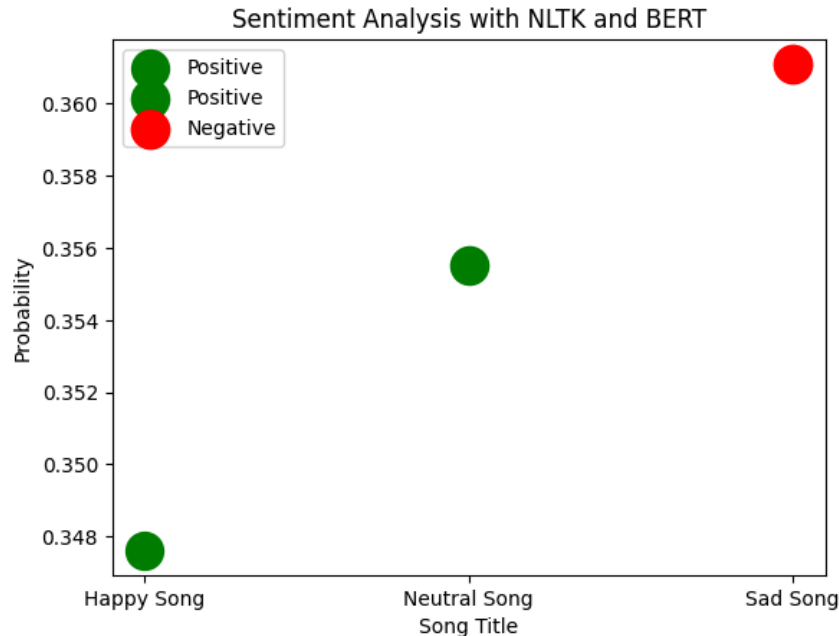


The graphical representation displayed above illustrates the performance of the vanilla BERT model in gauging sentiment. Notably, the graph exclusively depicts positive sentiment, revealing that, among the examined songs, "Happy" by Pharrell Williams exhibits a higher positive sentiment compared to the negative sentiment portrayed in "Traitor" by Olivia Rodrigo. However, a critical observation unveils a seemingly inaccurate depiction of sentiment, as the neutral song—a children's number song—registers the highest positive sentiment. This unexpected outcome prompts a deeper examination of the model's

proficiency in discerning nuances within sentiment categories. While the positive and negative songs align with their expected sentiment labels, the disproportionate elevation of positive sentiment for the neutral song raises questions about the model's capacity to accurately discriminate nuanced emotional tones within diverse musical genres. This intriguing revelation underscores the need for further investigation and refinement, urging us to delve into the intricacies of sentiment analysis to ensure a more precise and comprehensive understanding of the model's performance across varying emotional contexts.



The visual representation presented above delineates the performance of the BERT model augmented with a Recurrent Neural Network (RNN)/Long Short-Term Memory (LSTM) layer, featuring the same set of songs as previously analyzed. Noteworthy differentiations emerge when compared to the vanilla BERT results. Specifically, the sentiment graph now accurately reflects the neutral sentiment for the children's number song, aligning with our expectations based on the song's inherently neutral content. Moreover, the model adeptly captures the negative sentiment within "Traitor" by Olivia Rodrigo, indicating a commendable precision in discerning the melancholic tone embedded in the lyrics. However, a nuanced challenge emerges in the interpretation of the positive song, "Happy" by Pharrell Williams. Contrary to expectations, the sentiment graph registers a neutral sentiment for this ostensibly upbeat track. This divergence prompts a closer examination of the layered model's dynamics, highlighting its proficiency in capturing nuanced emotional nuances within varying song genres. The incorporation of the RNN/LSTM layer evidently contributes to a more refined sentiment analysis, particularly in instances where vanilla BERT falls short. The layered model's ability to accurately identify the neutral sentiment in the children's song and the negative sentiment in the ballad attests to its improved discriminative prowess. This nuanced performance underscores the enhanced capabilities conferred by the strategic integration of RNN/LSTM layers, demonstrating their efficacy in refining sentiment analysis within the intricate landscape of diverse musical expressions.



The graphical representation above elucidates the performance of the BERT model augmented with the Natural Language Toolkit (NLTK), employing the same set of songs for analysis. Notably, the sentiment graph indicates a commendable precision in identifying the negative sentiment within the melancholic composition "Traitor" by Olivia Rodrigo. Furthermore, the NLTK-layered model aptly discerns the positive sentiment in "Happy" by Pharrell Williams, aligning with the expected emotional tone of the song. However, a nuanced challenge surfaces as the sentiment graph registers a positive sentiment for the neutral children's number song, deviating from the anticipated neutral categorization. Despite this deviation, a comparative analysis suggests that the NLTK-layered model outperforms its RNN/LSTM-layered counterpart, particularly in accurately capturing the positive sentiment in the celebratory track and the negative sentiment in the somber composition. This differential proficiency prompts a closer examination of NLTK's unique strengths in linguistic processing, potentially enabling it to better navigate the subtleties of sentiment analysis. While neither model achieves perfection, the NLTK's more accurate depiction of sentiments suggests a nuanced advantage in leveraging linguistic features and contextual understanding. This observation underscores the iterative nature of sentiment analysis refinement and positions the NLTK-layered BERT model as a notable improvement over the vanilla BERT, offering a more nuanced and accurate interpretation of emotional nuances within the diverse musical landscape explored in our project.

In conclusion to this section, our experimentation with various models, particularly the NLTK-layered BERT, proved to be the most effective in discerning sentiment within song lyrics. However, it is essential to note that commendable results were achieved across all implementations, showcasing the versatility and adaptability of the employed methodologies. Despite minor variations in accuracy among the models, the overarching success lies in our ability to establish a meaningful connection between lyrics and mood. The incorporation of artificial intelligence (AI) allowed us to delve into the intricate relationship between lyrical content and emotional tone, providing valuable insights into the nuanced landscape of sentiment within musical compositions. While the NLTK-layered model emerged as the most proficient, each approach contributed to our understanding of sentiment analysis within the context of song lyrics. This

collective success underscores the feasibility of employing diverse methodologies to unravel the intricate interplay between music and emotions, ultimately enriching our comprehension of the complex and multifaceted nature of musical expressions.



Lyric Sentiment Classification By Artists

The model used in this section to classify the lyric sentiments is the transformer pipeline ‘j-hartmann/emotion-english-distilroberta-base.’ This model outputs a sentence’s sentiment score distributed between 7 different labels: joy, sadness, anger, fear, surprise, disgust, and neutral.

How would one go about classifying the general mood of an artist's lyrics? What would be the best approach?

A curated dataset comprising lyrical content from a diverse array of artists was constructed to facilitate the exploration of textual patterns and thematic elements inherent in songs. The compilation of artists in this study encompasses a wide range of musical styles, featuring notable figures such as Adele, Nirvana, Amy Winehouse, Dolly Parton, Lil Wayne, and Bob Marley. Additionally, the inclusion of Emily Dickinson, a revered poet, aligns with the growing perspective that regards song lyrics as a form of poetry, thereby enriching the breadth and depth of our textual analysis. In the construction of our dataset, each artist's lyrical content is arranged by songs, preserving the inherent narrative and thematic flow, except at the junctures where one song transitions to the next.

Given this structured compilation, a pivotal question arises regarding the optimal approach for sentiment analysis: whether to classify the lyrics of each song individually, subsequently aggregating these scores to ascertain the dominant sentiment, or to adopt a strategy of random sampling across an artist's lyrical repertoire to glean a general sentiment.

Comparing the different approaches:

Classifying Each Song Separately and Averaging the Sentiment:

Pros	Cons
Contextual Integrity: Analyzing entire songs maintains the lyrical and emotional context. Sentiments often develop or change throughout a song, and considering the whole song can capture this progression.	Computational Intensity: Analyzing entire songs is more computationally intensive, especially for artists with lengthy songs.
Song-Specific Insights: This method allows for song-specific analysis, which can be insightful for understanding the sentiment of particular pieces.	Potential Bias: If some songs have stronger sentiments than others, they could disproportionately influence the average sentiment.

Consistency Across Artists: If all artists' songs are treated similarly, comparisons across different artists become more meaningful.	
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Sampling Random Lyrics:

Pros:	Cons:
Efficiency: This method is less computationally demanding and faster, as it involves processing fewer lines.	Loss of Context: Random lines might lose the emotional and narrative context, leading to potential misinterpretation of sentiments.
General Sentiment Overview: It provides a broad, general overview of an artist's sentiment, which can be useful for a quick assessment.	Potential Skewness: Random sampling might unintentionally favor certain songs or types of lyrics, especially if some songs have more emotionally charged or neutral lines than others.

Average Sentiment by "Song" Approach

In the course of compiling the artist data for our sentiment analysis study, it was clear that our dataset consolidates lyrics without clearly distinguishing individual songs. The prospect of manually segmenting each text file into distinct songs presented itself as a considerably laborious and time-intensive task. To combat this challenge and devise a pragmatic approach, we sought guidance from ChatGPT regarding the average number of lyric lines typically found in a song. Acknowledging the inherent variability in song structures, we adopted an average count of 42 lines per song as a standard metric for our research purposes. This

The average number of lines in a song can vary significantly based on the genre and the artist's style. However, for a rough estimate:

- Verses:** A typical song has 2 to 4 verses. Each verse can have around 4 to 8 lines.
- Chorus:** The chorus, which is usually repeated several times throughout the song, might have around 4 to 8 lines.
- Bridge:** Not all songs have a bridge, but when they do, it's often 4 to 8 lines.

Adding these together, a rough estimate for an average song could be:

- 3 verses $\times$ 6 lines/verse = 18 lines
- 1 chorus $\times$ 6 lines = 6 lines (repeated 2-3 times, so 12-18 lines)
- 1 bridge $\times$ 6 lines = 6 lines

This gives a total of approximately 36 to 42 lines per song, not accounting for repetition of the chorus. Again, this is a very rough estimate and actual song lengths can vary widely.

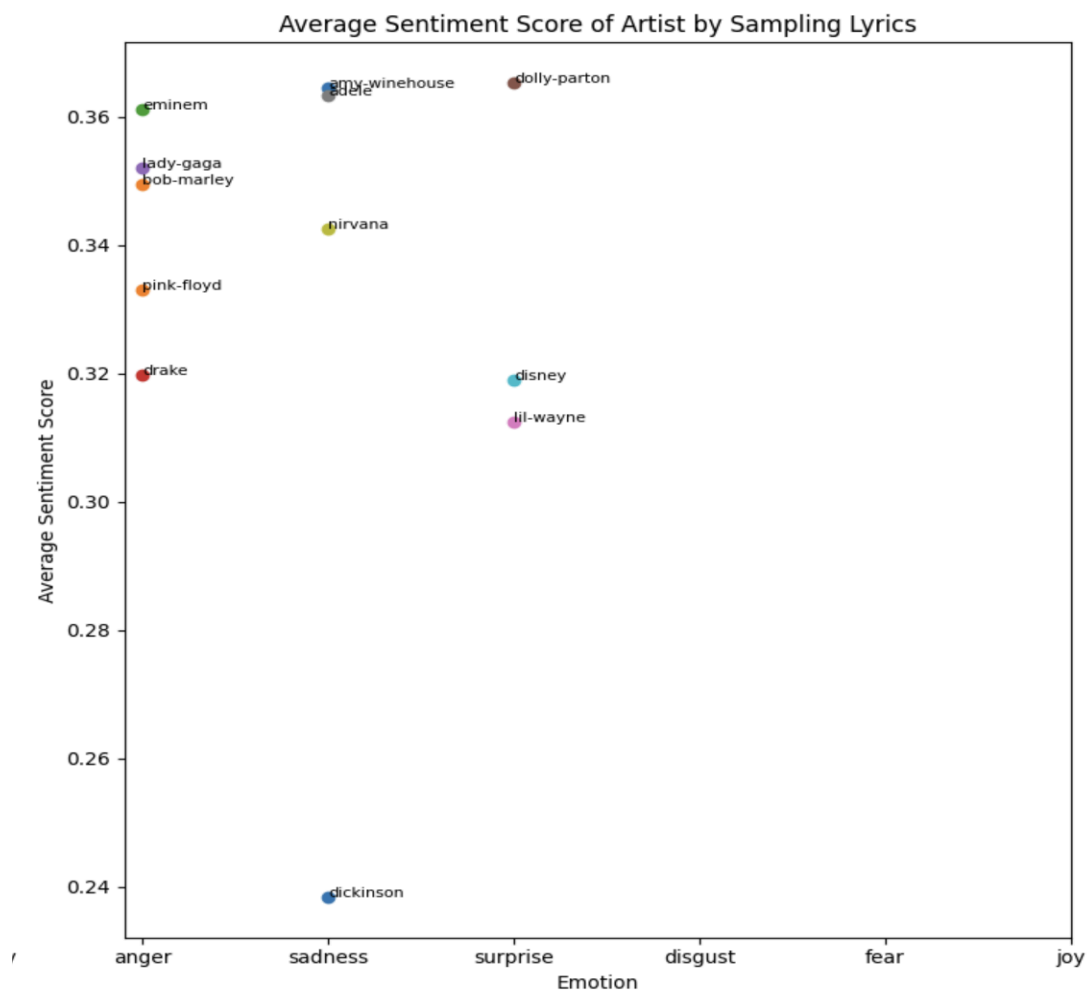
ChatGPT's response when prompted with "what is the average number of lyric lines in a song?"

decision allowed us to proceed with a more structured analysis while maintaining a balance between accuracy and feasibility in data segmentation.

Average Sentiment by Sampling Approach

To ensure consistency across the dataset, each text document, representative of an individual artist's lyrical oeuvre, was standardized to encompass a sample size of precisely 1500 lines. This normalization was executed regardless of the original volume of lyrical content, which ranged from around 1500 to 5000 lines per artist. The standardization process enabled a more equitable comparison and analysis across the corpus, ensuring that the results of the study were not biased by the length of the documents but rather reflective of the stylistic and thematic nuances of the artists' work.

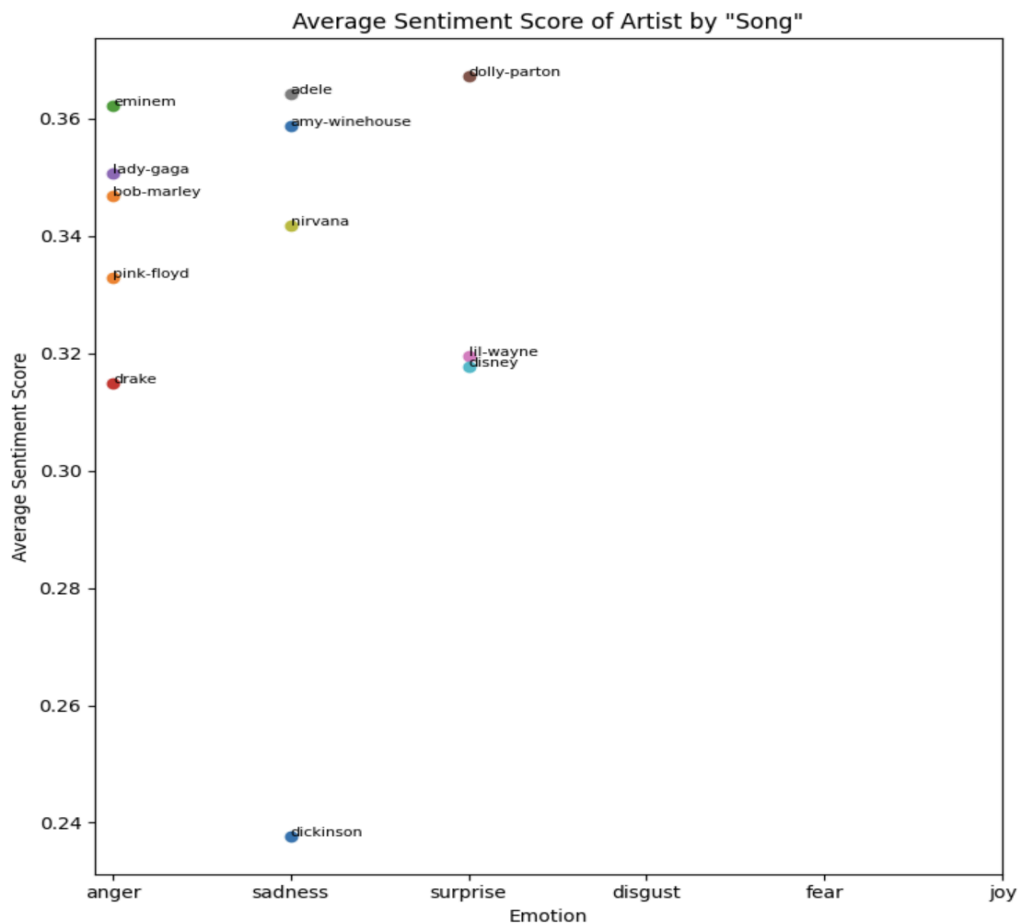
Results:



This diverse group of artists reveals intriguing results, particularly in the context of the emotional undertones associated with their lyrics. Notably, a predominant trend towards sad and angry sentiments is observed, which presents several unexpected insights. For instance, the lyrical content of Bob Marley,

traditionally associated with more uplifting themes, exhibits a significant degree of anger. Similarly, lyrics from Disney songs, often perceived as light-hearted, are categorized under "surprising," a classification that could potentially align more accurately with "exciting," albeit further investigation is required to substantiate this hypothesis.

Within this spectrum of sentiments, certain artists conform to anticipated patterns. Adele and Amy Winehouse, for example, are known for their emotive and often melancholic musical styles. The analysis interestingly suggests that while Adele's songs exhibit a higher intensity of sadness, the overall lyrical composition of Amy Winehouse's work is marginally more imbued with sadness compared to Adele's. This nuanced differentiation in the sentiment profiles of their lyrics offers a compelling perspective on the emotional depth and complexity characterizing their musical expressions.



As we can see, there is not much of a difference in average artist sentiment scores between methods of processing the lyrics. Not to mention, the scores are all on the lower side(<0.4).

Fine Tuning with Negation

Does this pre-trained model give accurate results? This task is complicated by linguistic nuances, such as the use of negation, which can significantly alter the sentiment conveyed by the lyrics. Negation handling is a critical aspect of natural language processing that requires careful attention to detail, as it can reverse the polarity of an otherwise straightforward sentiment expression.

For instance, the lyrics of Bob Marley, often imbued with positive and uplifting messages, might be misclassified as 'angry' by transformer models that do not adequately account for negation. For example, take one of Marley's classic lines, "Don't worry about a thing," and look how the model interprets its sentiment:

```
Don't worry about a thing
Top Sentiment: {'label': 'fear', 'score': 0.887710452079773}
```

In our study, we identified a critical shortcoming of this sentiment analysis model: its inability to accurately interpret negated constructs, such as "Don't worry," which was inaccurately labeled with 'fear'. Addressing this issue was an important objective of our research, leading us to develop a negation handling function. This function analyzes the input sentence and, upon detecting words that imply negation, outputs a semantically altered version of the sentence where negated words are replaced with their antonyms.

Our methodology draws inspiration from the work of Izunna Okpala et al. in "A Semantic Approach to Negation Detection and Word Disambiguation with Natural Language Processing." According to their algorithm, the process entails iterating through each word in a sentence to identify 'negation flags'—words that indicate negation, such as 'not' or 'don't.' Following the detection of these flags, the approach involves substituting the negated term with its opposite meaning, or antonym, to reflect the true sentiment more accurately. This technique enhances the model's capability to comprehend and process negated phrases, thus improving the accuracy of sentiment analysis in lyrical content.

Algorithm 2 Detect Negation

```
procedure DETECT_NEG(text)
  ref ← ["not", "nor", "never", "neither"]
  text ← text.split()
  for ref IN text do
    if (any item IN ref) then
      st = set(ref)
      if (e IN st) then
        for i, e IN enumerate(text) do
          return i
        end for
      end if
    end if
  end for
end procedure
```

Algorithm 4 Word Disambiguation

```
procedure DISAMB(sentence)
  token ← nltk.word_tokenize(sentence)
  tags ← nltk.pos_tag(token)
  if tags[found] = NEGATED then
    token[int_index + 1] = get_antonyms(token)
    token.pop(int_index)
    return token
  else
    return token
  end if
end procedure
```

Our function integrates the basis of Okpala et al.'s proposed algorithms for transforming sentences. This function utilizes the Natural Language Toolkit (NLTK) WordNet library to identify antonyms for negated constructs within a sentence. Moreover, it incorporates a mechanism to handle intensifiers such as "very" and "really," ensuring that the negation of the sentiment is accurately maintained. The selection of the most appropriate antonym from a list is informed by the frequency distribution within the Brown corpus, which is employed to identify the 'top common word.'

The effectiveness of this approach was evident in the preprocessing of lyrics before their sentiment classification. Despite recognizing that our negation handling method is not without flaws and may occasionally deviate from the intended sentiment of the sentence, it has shown promising results.

OG Sentence: I just can't be without you Emotion: sadness

Modded Sentence: I just go without you Emotion: neutral

For example, the sentence "I just can't be without you," originally classified as sad, shifted to neutral after negation handling. This highlights a potential area for refinement in our model.(And of course, using more data)

Here is a case where the negation handling function appears to help make the model classification more accurate:

```
OG Sentence:  Hey, lover, sugar don't you see? Emotion:  anger
Modded Sentence:  Hey, lover, sugar me see? Emotion:  joy
```

A sentence laden with ostensibly joyful words like 'lover' and 'sugar' was initially overpowered by the presence of 'don't,' leading to an 'anger' sentiment classification. However, after the application of our negation handling, the true sentiment was more faithfully represented.

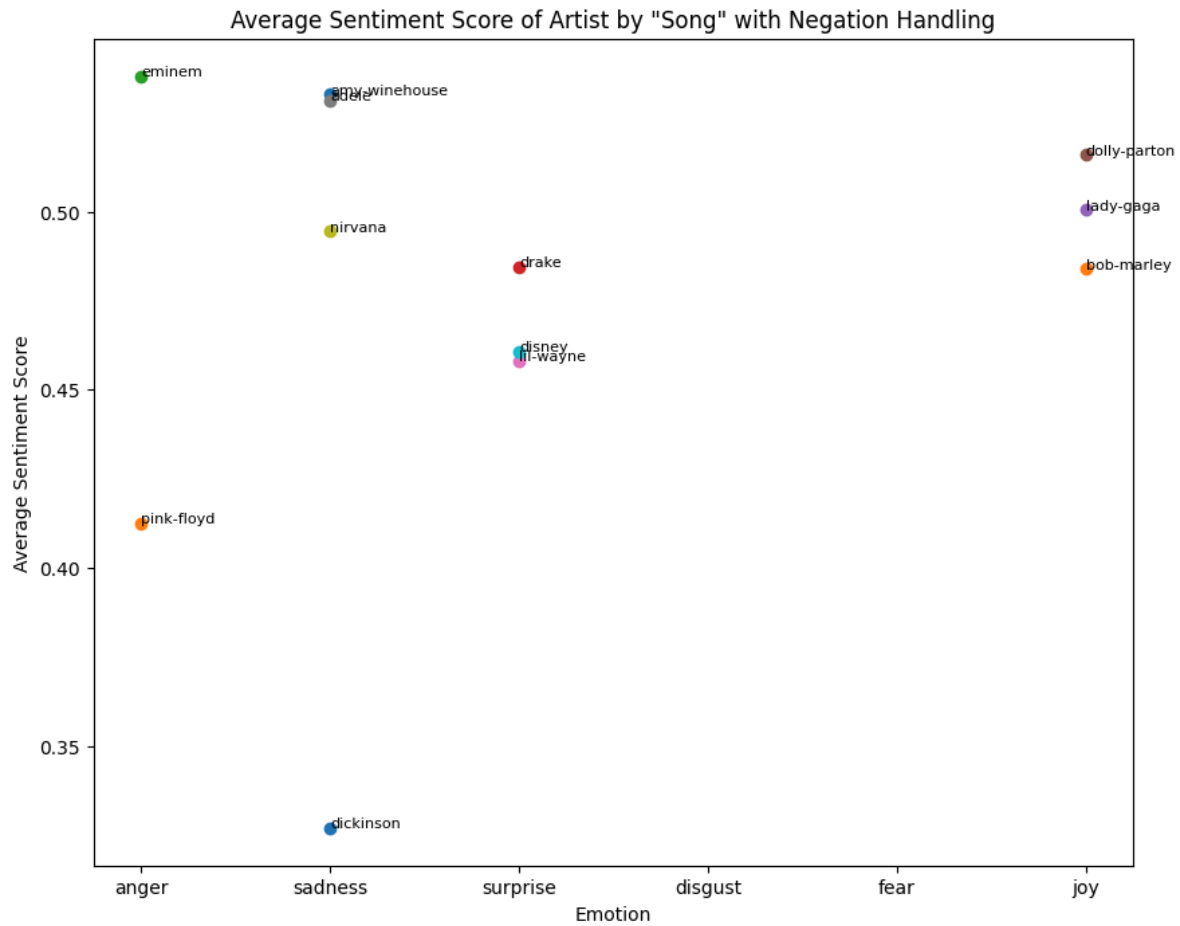
Going back to Bob Marley's lyric, "Don't worry about a thing," this was the sentiment given to its modified version:

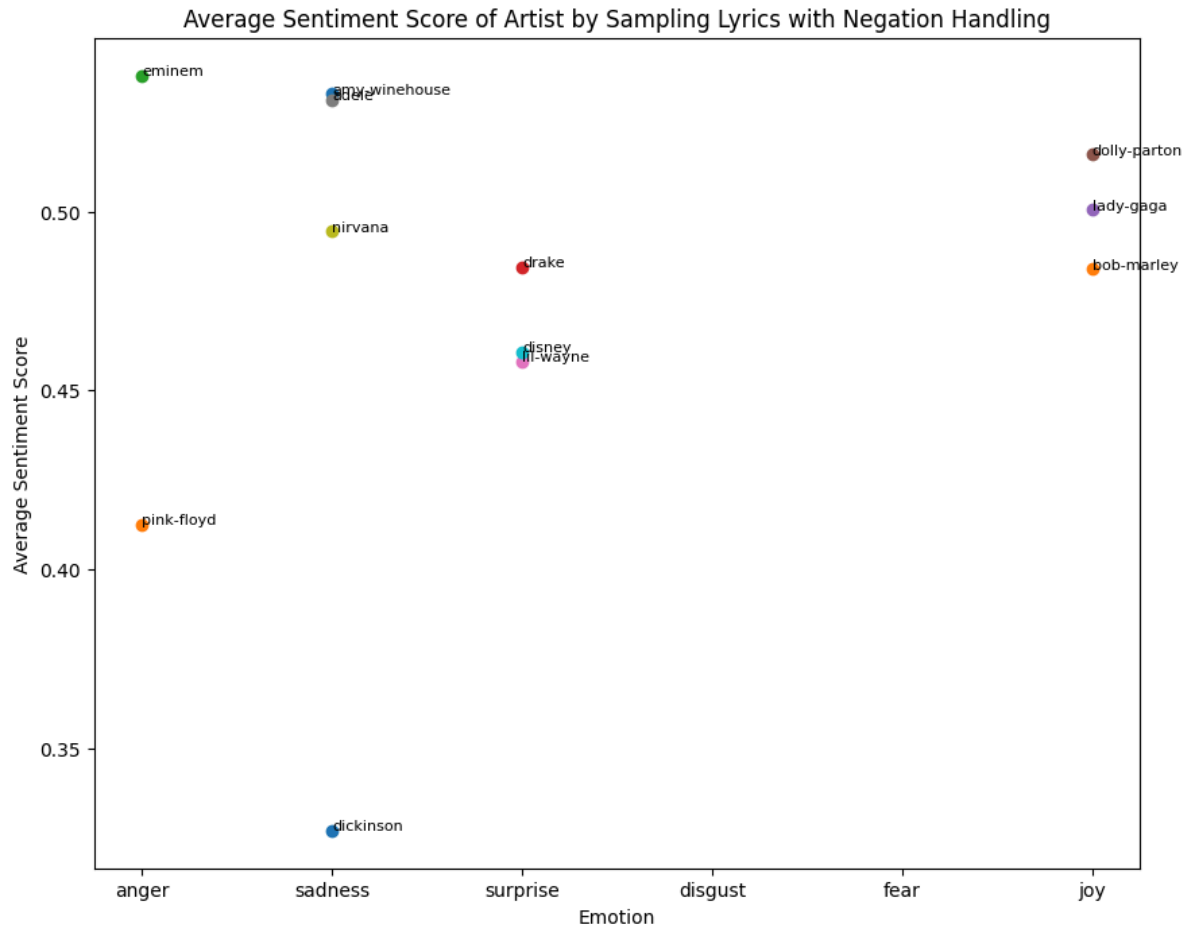
```
Original Sentence:
Don't worry about a thing
Top Sentiment:  {'label': 'fear', 'score': 0.887710452079773}

Modified Sentence:
Reassure about a thing
Top Sentiment:  {'label': 'neutral', 'score': 0.6557764410972595}
```

Although not perfect, this lyric post-negation handling resonated more closely with the song's positive essence. This underscores the importance of effective negation handling in sentiment analysis and its impact on model accuracy. Our ongoing efforts include exploring alternative corpora that may provide a more robust common antonym selection, handling double negatives and negation intensity, all with the goal of further refining our approach and enhancing the precision of our sentiment analysis model.

After preprocessing the lyrics with the negation handling function, we then reevaluated the average top sentiment score for the example artists:



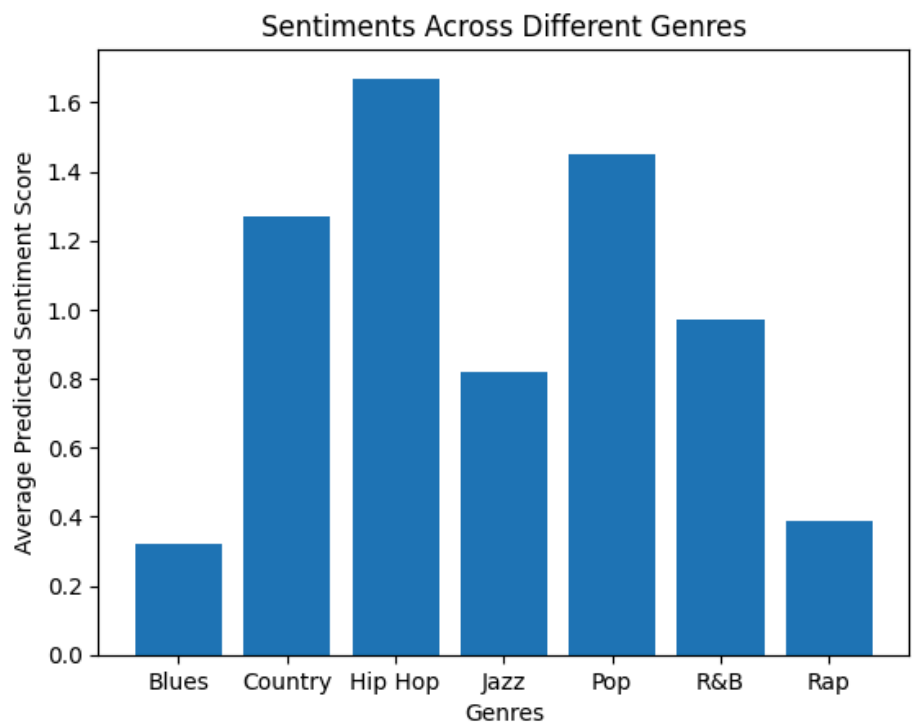


The outcomes of our sentiment analysis indicate notable shifts in the emotional classification for certain artists. The implementation of our negation handling function has resulted in a transition from sentiments of 'anger' to 'joy' for artists like Bob Marley, Lady Gaga, and Dolly Parton. These updated classifications arguably offer a representation that aligns more closely with the general perception of these artists' musical expressions. Furthermore, the study reaffirms that the choice of sampling methods exerts minimal influence on the overall results, suggesting the robustness of our sentiment analysis approach in capturing the prevailing emotional tone of an artist's body of work.



By Genre

Our exploration into the diverse realms of musical moods and emotions, dissected through the lens of various genres, constitutes a multifaceted journey in which we navigated the vast landscape of musical expression. Acknowledging the inherent broadness of genres as a unit of measurement, we embarked on the ambitious task of deciphering overarching trends that shed light on the predominant sentiments within different musical categories. Recognizing that genres serve as broad strokes to categorize an extensive range of sonic expressions, we approached our analysis with the awareness that the musical intricacies within each genre are as diverse as the genres themselves. Despite this inherent complexity, we deemed it valuable to extrapolate and explore the general emotional inclinations inherent in different genres, understanding that such overarching claims would necessarily lack the granularity to capture every nuance. Armed with advanced natural language processing (NLP) models, including the formidable BERT, we traversed the musical landscape, uncovering nuanced patterns and sentiments interwoven within lyrics. The integration of our NLP models allowed us to distill general trends that transcended the complexities of individual songs and artists, providing a macroscopic perspective on the emotional undercurrents prevalent in various genres. These findings, while acknowledging the inherent limitations of broad categorizations, proved to be not only insightful but also intriguing, offering a glimpse into the overarching emotional tapestry that different genres contribute to the collective musical experience. In navigating the amalgamation of data and diverse musical expressions, our methodologies and models facilitated the extraction of meaningful insights, underscoring the interdisciplinary synergy between music, natural language processing, and artificial intelligence. Through this exploration, we contribute to the ongoing dialogue surrounding the symbiotic relationship between music and emotion, unraveling layers of complexity to discern broad emotional trends that resonate across the diverse spectrum of musical genres.



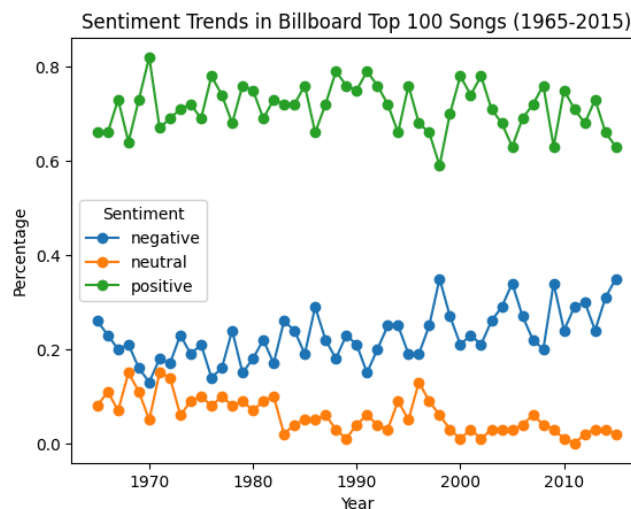
The bar graph presented above encapsulates a panoramic view of the prevailing sentiments within various music genres, offering a compelling snapshot of the emotional tapestry inherent in musical expressions. Our methodology for constructing this comprehensive analysis involved the meticulous input of lyric data from diverse genres, leveraging the analytical capabilities of BERT and other natural language processing (NLP) models to extract nuanced sentiments. This intricate process necessitated an initial data cleaning phase, a meticulous curation aimed at ensuring the integrity and relevance of the dataset. The preliminary cleaning steps were pivotal, guaranteeing that the subsequent sentiment analysis accurately reflected the intended emotional nuances encapsulated within the lyrics. As our models delved into the diverse musical genres, ranging from the rhythmic beats of pop to the soulful depths of blues, the sentiment analysis became a powerful lens through which we could discern general trends in emotional inclination. The resultant bar graph, a visual manifestation of this computational odyssey, not only captures the overarching emotional tendencies within each genre but also serves as a testament to the interdisciplinary synergy between music, data science, and artificial intelligence. This methodological fusion facilitated the extraction of meaningful insights, transcending the surface-level examination of genres to unravel the complex interplay between lyrical content and emotional resonance. Thus, our analytical endeavor stands as an illuminating contribution to the broader discourse surrounding the intrinsic connection between music and sentiment, illustrating how advanced computational techniques can unveil hidden patterns within the rich and diverse fabric of musical genres.

The numerical representation employed in our sentiment analysis, where 0 denotes negativity, 1 signifies neutrality, and 2 represents positivity, serves as a quantifiable framework that allows for a systematic evaluation of emotional tones within diverse music genres. This numerical scale not only provides a structured basis for categorizing sentiments but also encapsulates the complexity of emotional nuances within the lyrical content of songs. An intriguing revelation surfaces as we observe the distribution of sentiments across various genres, wherein a noteworthy trend emerges—most genres exhibit a proclivity towards negative sentiments. This observation prompts a profound contemplation on the interplay between musical expression and emotional resonance, suggesting that a multitude of genres may inherently gravitate towards exploring themes and narratives that encapsulate a range of melancholic, contemplative, or introspective sentiments. This finding, while capturing a broad overarching trend, underscores the multifaceted nature of musical genres, each serving as a unique vessel for emotional expression. The numerical framework, in this context, becomes a valuable tool for distilling and quantifying the intricate emotional landscape of music, revealing patterns that invite further exploration and contextualization within the broader cultural and artistic tapestry. The prevalence of negative sentiments across genres beckons a deeper inquiry into the socio-cultural and psychological dimensions that shape musical expressions, thereby contributing to a richer understanding of the profound interplay between music and human emotions.

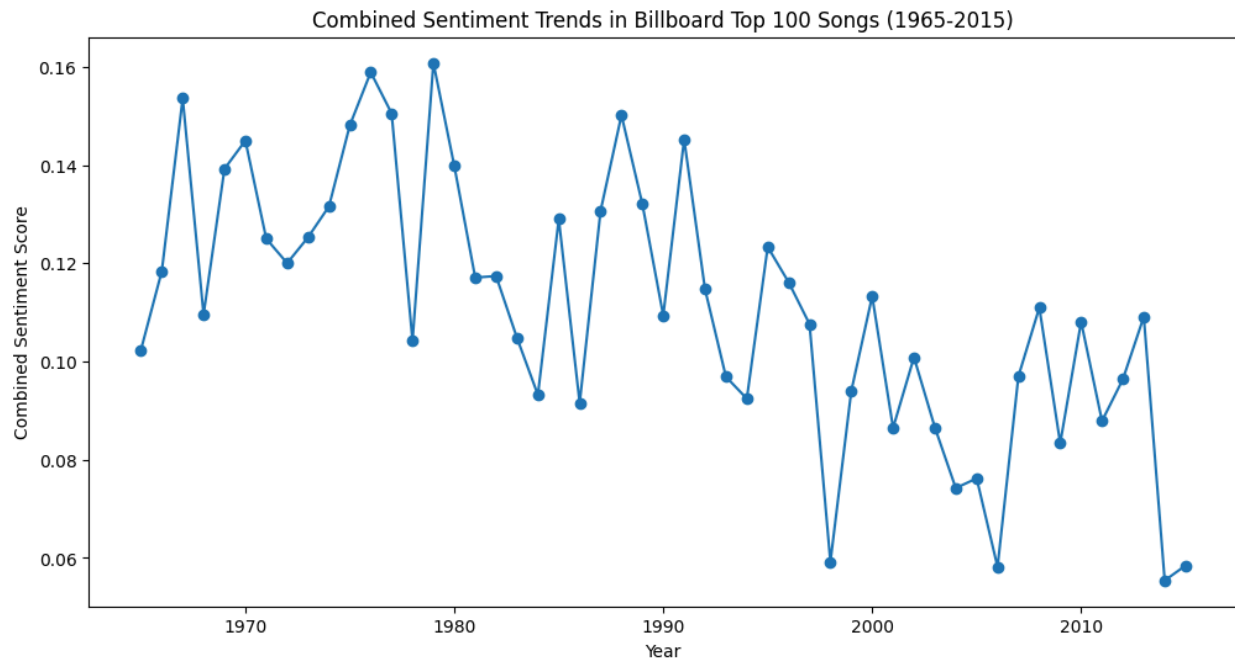
Trends over time

We collectively embarked on the task of uncovering fascinating insights into the emotional trends embedded within the lyrics of songs that have graced the prestigious Billboard Top 100 charts since 1965. Our first crucial step involved meticulous data preprocessing using the pandas library in Python. This phase included loading the dataset, handling missing values, and extracting relevant columns such as 'Song,' 'Artist,' 'Year,' and the lyrical content itself. Subsequently, armed with the TextBlob library, we delved into the sentiment analysis of the lyrics. Each song's sentiment—whether positive, negative, or neutral—was discerned, attributing a sentiment score to encapsulate the emotional tone of the lyrics.

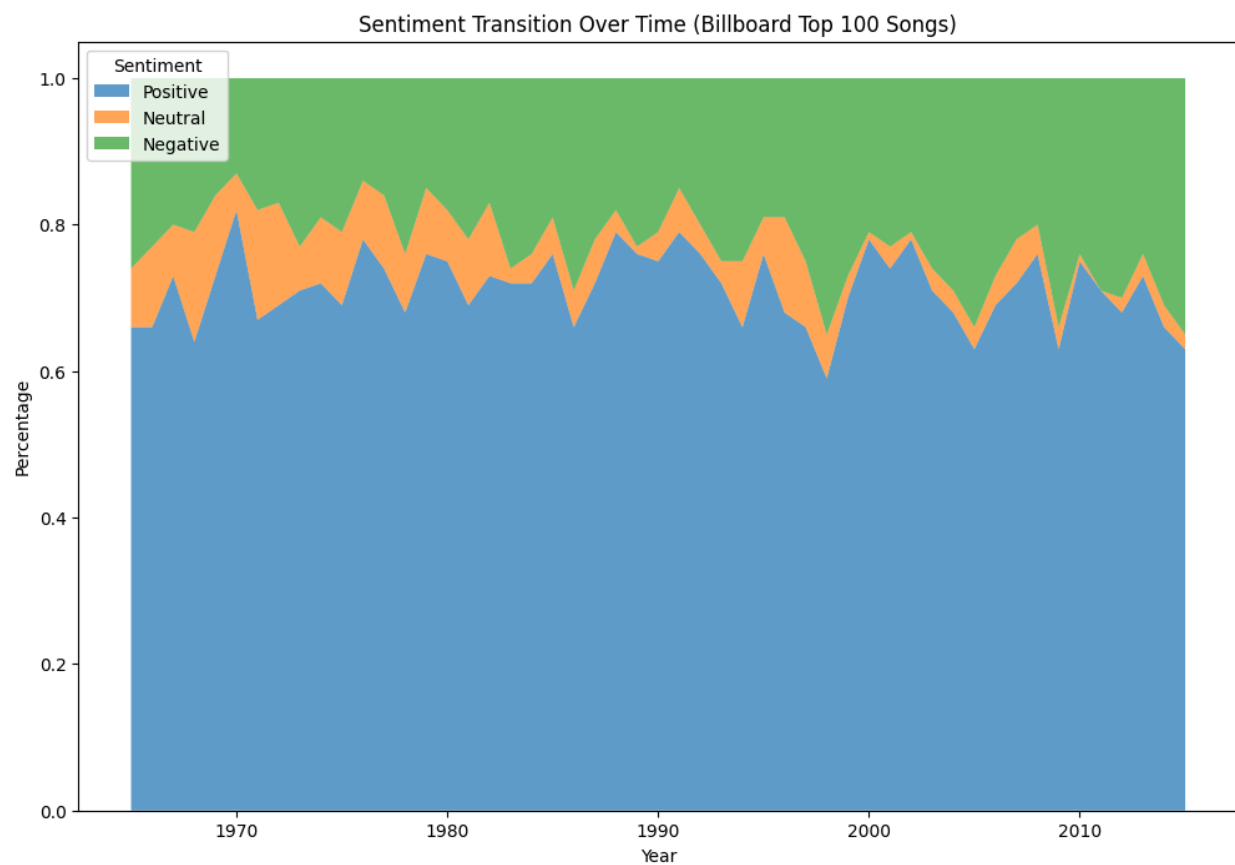
In this segment of the project, we focused on loading and preprocessing the Billboard Top 100 dataset, a crucial foundation for the subsequent sentiment analysis. Leveraging the pandas library in Python, we imported the dataset from the specified file path, addressing encoding nuances using the 'latin-1' encoding to ensure accurate data ingestion. Following this, the sentiment analysis unfolded through the implementation of the TextBlob library. A sentiment analysis function was devised to evaluate the emotional polarity of song lyrics, categorizing them as 'positive,' 'negative,' or 'neutral' based on their sentiment scores. This sentiment classification was then applied to the lyrics column of the dataset, creating a new 'Sentiment' column to capture the emotional essence of each song.



This analytical step involved aggregating this wealth of emotional data by year, providing a nuanced perspective on how the collective sentiment of Billboard's chart-topping songs has transformed over the decades. Leveraging the pandas library, the data was grouped by year, and the mean sentiment score for each year was calculated. This meticulous aggregation process laid the foundation for a compelling visual narrative—a line graph illustrating the trajectory of combined sentiment scores from 1965 to 2015. The resulting graph, generated using the Matplotlib library, serves as an insightful depiction of the evolving emotional landscape embedded within the lyrics of Billboard Top 100 songs, offering a tangible representation of the changing sentiments that have resonated through the years.



Heat map of the sentiment analysis:



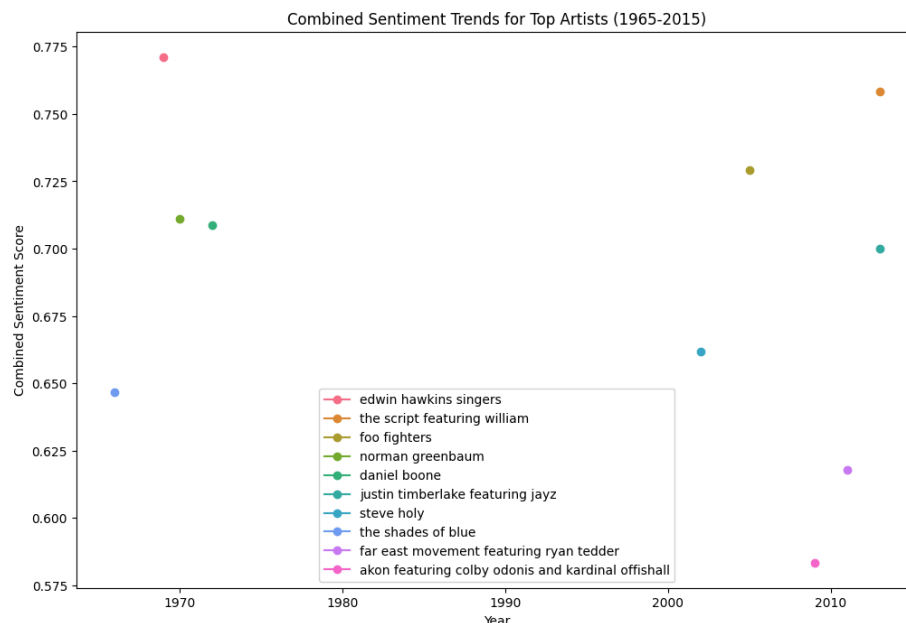
We delved into a more nuanced analysis of song lyrics, leveraging the CountVectorizer module from scikit-learn—a technique we've explored in our NLP class. To ensure the integrity of the original data, we

created a duplicate dataset, `df_copy`, addressing potential gaps in the 'Lyrics' column by filling NaN values with empty strings. Employing `CountVectorizer`, we extracted bigrams, consecutive pairs of words, revealing intricate linguistic patterns within the lyrics. This approach aligns with our classroom learnings on text vectorization. The resulting bigram matrix was transformed into a `DataFrame`, named `bigram_df`, facilitating a detailed examination of prevalent two-word combinations in Billboard Top 100 songs. To distill meaningful insights, we sorted the `DataFrame` by bigram counts in descending order, extracting and displaying the top 10 significant bigrams. This comprehensive analysis adds depth to our understanding of the textual fabric within the dataset, showcasing recurrent linguistic pairs that contribute to the rich tapestry of language in the lyrics of chart-topping songs over the years.

The results were:

	Count
oh oh	3075
yeah yeah	1734
dont know	1724
la la	1560
na na	1366
love love	1165
im gonna	1092
dont want	841
baby baby	813
dont wanna	798

In this final analytical phase, the focus shifts to a targeted examination of sentiment trends over time for specific artists within the Billboard Top 100 dataset.

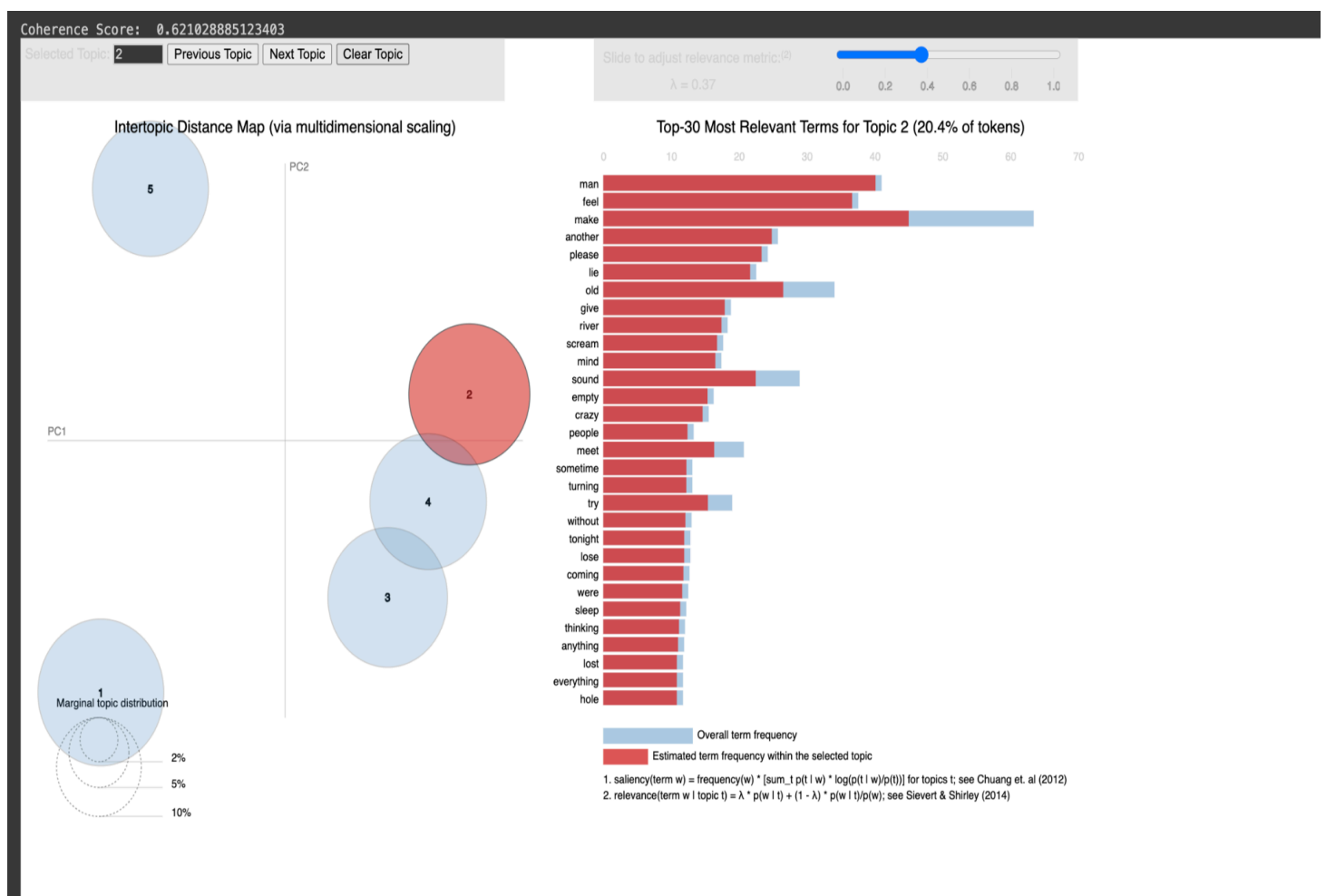


Extended Analysis

To enhance our understanding of thematic structures and meaning within artists' lyrics, we have conducted an experiment using Gensim's pyLDAvis LDA Model for topic modeling. Our objective was to integrate the insights garnered from LDA model analysis with those derived from sentiment analysis, as the combination of the two methodologies often yields a comprehensive depiction of textual nuances. The convergence of LDA models, which excel at discerning clusters of frequently co-occurring words, with sentiment analysis, capable of interpreting emotive undertones, equips us with a robust toolset for dissecting and interpreting the rich tapestry of themes woven into lyrical compositions.

Topic Modeling of Pink Floyd lyrics:

The visualization presents an intertopic distance map created via multidimensional scaling, where our lyrical data is categorized into five distinct topics. Topic 2, prominently highlighted, accounts for 20.4% of tokens and is depicted in close proximity to the other topics, indicating thematic similarities within the dataset. This proximity suggests a degree of overlap or relatedness in the content of the lyrics encompassed by these topics. The bar graph to the right details the top-30 most relevant terms for Topic 2, providing a glimpse into the thematic elements that characterize this particular cluster within the lyrical data. The coherence score at the top (~0.621) suggests a moderate level of consistency within the topics identified by the model.



If you recall from the graph of average sentiment scores for different artists, Pink Floyd's average sentiment was anger. This makes sense as they often write songs with themes of war, greed, political and social commentary. For example, the album *The Dark Side of the Moon*, centers on the concept of madness, exploring themes like conflict, greed, time, death, and mental illness. Upon reviewing the lyrical content of Pink Floyd as an illustrative example, it appears that Topic 2 prominently features words indicative of a theme centered around madness.



In contemplating our future research and analytical endeavors utilizing LDA modeling, the repository of resources at our command is substantial. Tools such as pyLDAvis offer a profound means for not only identifying but also endeavoring to categorize the thematic essence within the lyrics of various artists. Moreover, there lies the potential to refine an LDA model by leveraging a dataset of sentences that are annotated with explicit thematic classifications. Such advancements would undoubtedly accelerate the acuity of topic detection and thematic labeling, propelling the domain of lyrical analysis into new realms of precision and depth.



Here are the top words associated with Topic 2

Conclusion

In this exploration of sentiment analysis within the realm of song lyrics, our project seamlessly integrates advanced natural language processing techniques, ranging from the fine-tuning of transformer models like BERT to traditional NLP tools. Through meticulous data preprocessing and sentiment analysis, we unearthed not only temporal trends in emotional expression but also delved into linguistic nuances by exploring prevalent bigrams. The project's scope extended beyond individual songs, encapsulating a dynamic analysis of sentiment transitions over time and artist-specific trends. The visualizations crafted using Matplotlib vividly portray the intricate emotional landscapes woven into the lyrical fabric of diverse musical genres. This interdisciplinary journey not only contributes significantly to the field of sentiment analysis but also underscores the symbiotic relationship between technology and the intricate world of musical expression. The comprehensive insights gained from this project illuminate the multifaceted dimensions of sentiment analysis within the rich tapestry of song lyrics, demonstrating the potential for advanced computational techniques to unravel hidden patterns and nuances in the emotional language of music.

How successful we think we were

Reflecting on the entirety of our project, we consider our efforts to be largely successful in achieving the outlined objectives. Our meticulous application of sentiment analysis techniques, coupled with the integration of advanced natural language processing models, has provided a nuanced understanding of emotional trends within song lyrics. The visual representations, including line graphs depicting temporal shifts and stacked area plots illustrating sentiment transitions, offer compelling insights into the complex interplay between music and emotions. The successful fine-tuning of transformer models like BERT, alongside the exploration of linguistic patterns through bigrams, showcases the versatility and adaptability of our computational approaches. Additionally, the artist-specific analyses contribute to a richer narrative, portraying the unique emotional trajectories of individual musicians over time. While acknowledging the inherent challenges and nuances within sentiment analysis, we believe our project successfully navigates these complexities, offering a holistic exploration of emotional dynamics within the diverse landscape of song lyrics. The synthesis of technological advancements with artistic expression has yielded valuable contributions to the understanding of sentiment analysis in the musical domain, making our project a noteworthy and insightful endeavor.

Possible Further Exploration

As we conclude this project, it is evident that there exists a myriad of avenues for further exploration and expansion. One promising avenue is the refinement and augmentation of our sentiment analysis models to encompass a more extensive and diverse set of emotions, moving beyond the binary classification of positive, neutral, and negative sentiments. Exploring the complexities of nuanced emotions, such as nostalgia, excitement, or introspection, could lend a deeper layer of understanding to the emotional fabric woven into music. Additionally, delving into the impact of cultural and regional influences on musical sentiments could provide insights into the dynamic interplay between music and societal expressions. Moreover, extending our creative lyric generation capabilities to capture the distinctive styles of renowned artists within specific genres could elevate the model's creative potential. Collaborative efforts with musicians and lyricists could further enrich our project, fostering a symbiotic relationship between artificial intelligence and human creativity. Furthermore, integrating real-time data streams from social

media and music platforms could offer a dynamic perspective on contemporary emotional trends in the music landscape. Overall, the project's success serves as a springboard for future endeavors that explore the boundless intersections of technology, sentiment analysis, and creative expression within the vibrant tapestry of musical artistry. With additional time and resources, these avenues could be traversed to unveil even more profound insights and innovations at the confluence of artificial intelligence and musical exploration.

Division of work:

Mary: Did the layering of the RNN/LSTM and NLTK models on BERT and worked with vanilla BERT, found the sentiment analysis by genre, worked on writing up the final report.

Beck: Did analysis of lyrics by artist and fine tuning with negation handling. Also explored LDA models. Worked on writing up the final report

Lily: Did analysis of the trends of sentiment analysis over time. Worked on writing up the final report.

Sources:

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